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Bayes' Theorem

Suppose we have estimated **prior** probabilities for events we are concerned with, and then obtain new information. We would like to a sound method to computed *revised* or **posterior** probabilities. Bayes' theorem gives us a way to do this.

Probability using Bayes’ Theorem

Prior Probabilities → New Information → Application of Bayes’ Theorem → Posterior Probabilities

Application of Bayes' Theorem

- Consider a manufacturing firm that receives shipment of parts from two suppliers.
- Let A_1 denote the event that a part is received from supplier 1; A_2 is the event the part is received from supplier 2

We get 65 percent of our parts from supplier 1 and 35 percent from supplier 2.

Thus:

$P(A_1) = .65$ and $P(A_2) = .35$

Quality levels differ between suppliers

	Percentage Good Parts	Percentage Bad Parts
Supplier 1	98	2
Supplier 2	95	5

Let G denote that a part is good and B denote the event that a part is bad. Thus we have the following conditional probabilities:

$$P(G \mid A_1) = .98 \text{ and } P(B \mid A_2) = .02$$

$$P(G \mid A_2) = .95 \text{ and } P(B \mid A_2) = .05$$

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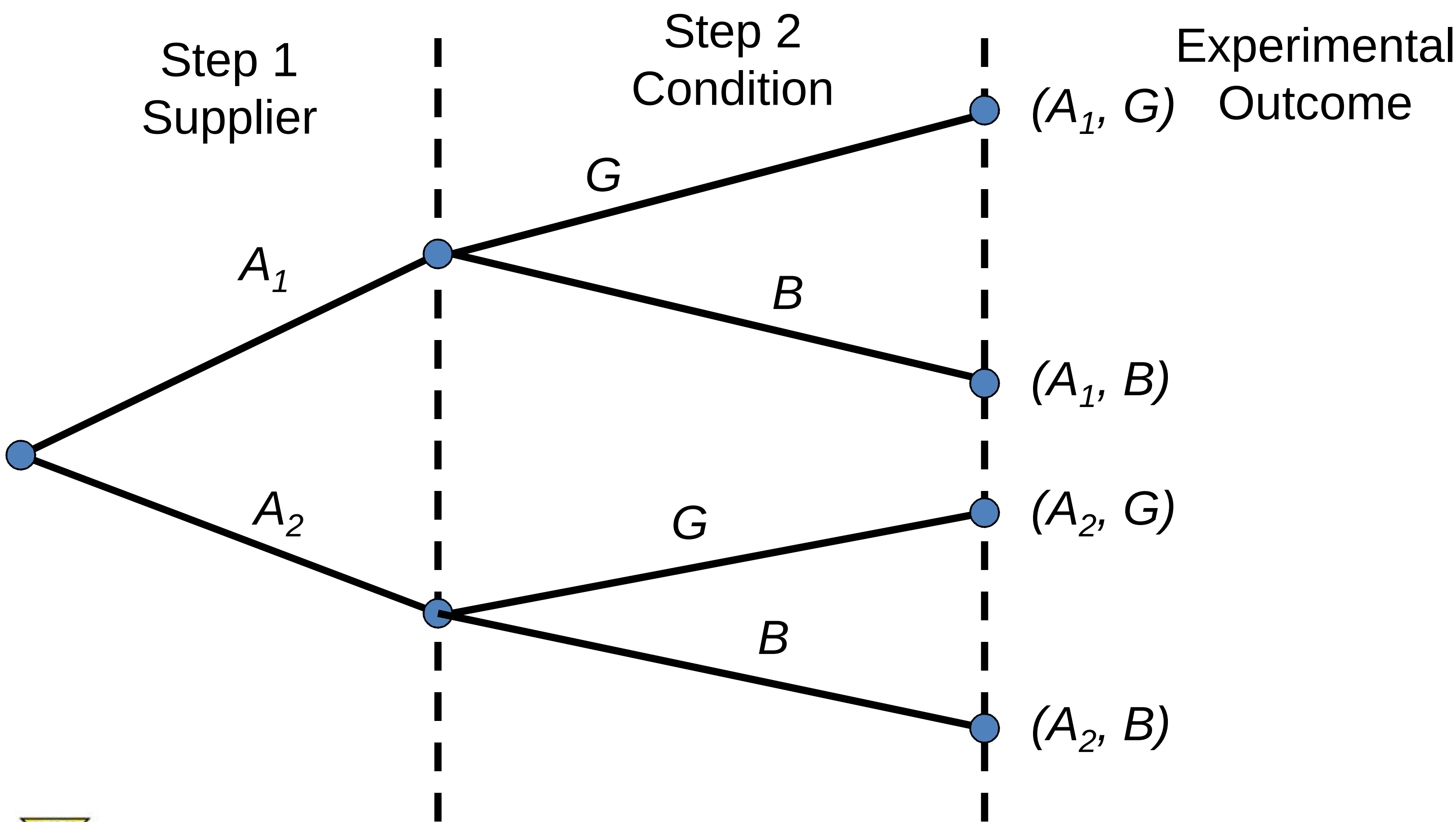
Tree Diagram for Two-Supplier Example`



Each of the experimental outcomes is the intersection of 2 events. For example, the probability of selecting a part from supplier 1 that is good is given by:

$$P(A_1, G) = P(A_1 \cap G) = P(A_1)P(G \mid A_1)$$

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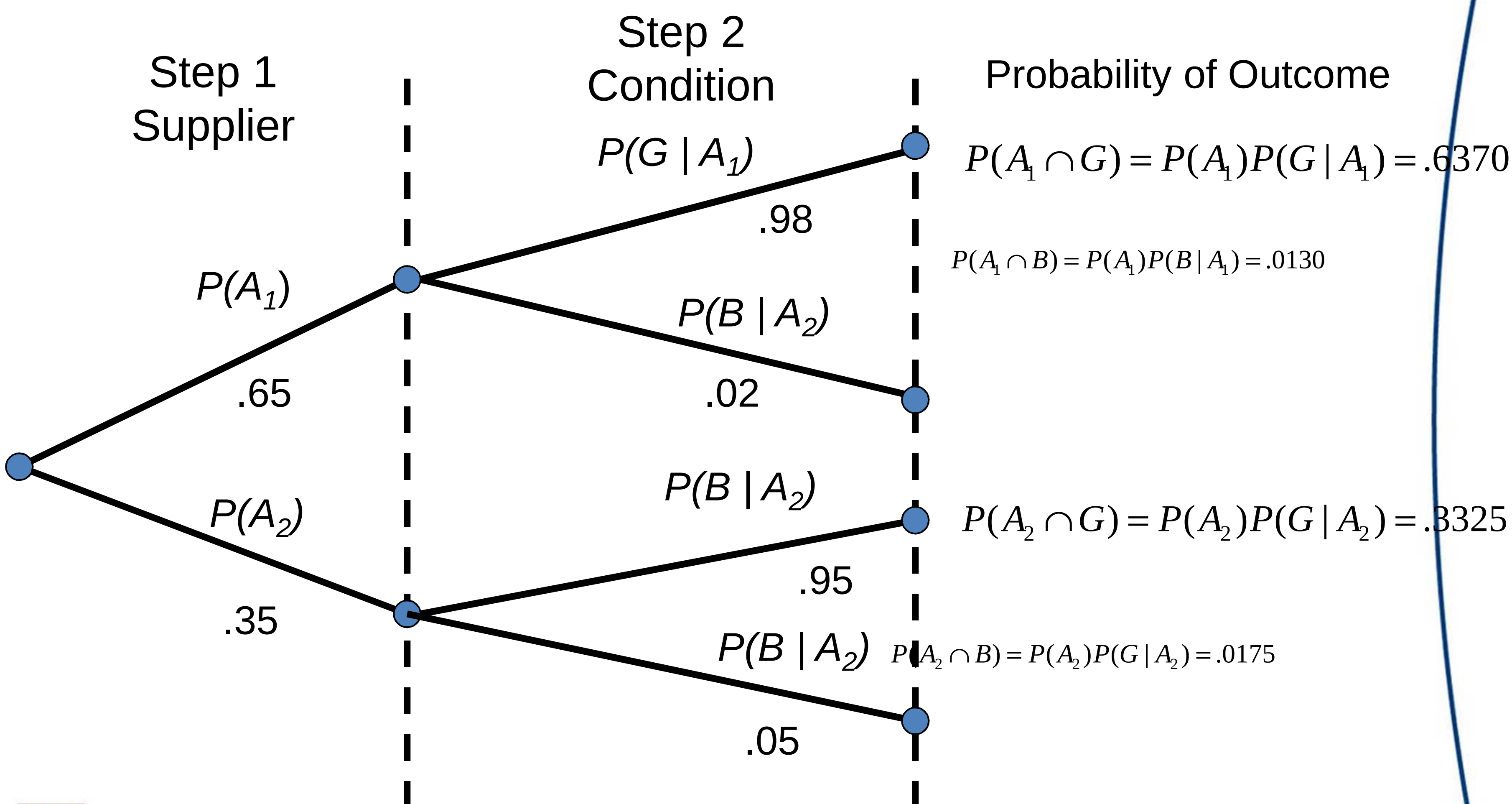
Probability Tree for Two-Supplier Example



A bad part broke one of our machines—so we're through for the day. What is the probability the part came from supplier 1?

We know from the law of conditional probability that:

$$P(A_1 \mid B) = \frac{P(A_1 \cap B)}{P(B)} \qquad (4.14)$$

Observe from the probability tree that:

$$P(A_1 \cap B) = P(A_1)P(B \mid A_1) \qquad (4.15)$$

The probability of selecting a bad part is found by adding together the probability of selecting a bad part from supplier 1 and the probability of selecting bad part from supplier 2.

That is:

$$\begin{aligned} P(B) &= P(A_1 \cap B) + P(A_2 \cap B) \\ &= P(A_1)P(B \mid A_1) + P(A_2)P(B / A_2) \end{aligned} \qquad (4.16)$$

Bayes' Theorem for 2 events

By substituting equations (4.15) and (4.16) into (4.14), and writing a similar result for $P(B \mid A_2)$, we obtain Bayes' theorem for the 2 event case:

$$P(A_1 \mid B) = \frac{P(A_1)P(B \mid A_1)}{P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2)}$$

$$P(A_2 \mid B) = \frac{P(A_2)P(B \mid A_2)}{P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2)}$$

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Do the Math

$$P(A_1 \mid B) = \frac{P(A_1)P(B \mid A_1)}{P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2)}$$

$$= \frac{(.65)(.02)}{(.65)(.02) + (.35)(.05)} = \frac{.0130}{.0305} = .4262$$

$$P(A_2 \mid B) = \frac{P(A_2)P(B \mid A_2)}{P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2)}$$

$$= \frac{(.35)(.05)}{(.65)(.02) + (.35)(.05)} = \frac{.0175}{.0305} = .5738$$

Bayes' Theorem

$$P(A_i \mid B) = \frac{P(A_i)P(B \mid A_i)}{P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2) + \ldots + P(A_n)P(B \mid A_n)}$$

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Tabular Approach to Bayes' Theorem—2-Supplier Problem

(1) Events A_i	**(2)** Prior Probabilities $P(A_i)$	**(3)** Conditional Probabilities $P(B \mid A_1)$	**(4)** Joint Probabilities $P(A_i \cap B)$	**(5)** Posterior Probabilities $P(A_i \mid B)$
A_1	.65	.02	.0130	.0130/.0305 =.4262
A_2	.35	.05	.0175	.0175/.0305 =.5738
	1.00		P(B)=.0305	**1.0000**

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Using Excel to Compute: Posterior Probabilities

Events	Prior Probabilities	Conditional Probabilities	Joint Probabilities	Posterior Probabilities			
A_1	0.65	0.02	0.013	0.426229508			
A_2	0.35	0.05	0.0175	0.573770492			
			0.0305	1.0000			

41. A consulting firm submitted a bid for a large consulting contract. The firm's management felt id had a 50-50 change of landing the project. However, the agency to which the bid was submitted subsequently asked for additional information. Past experience indicates that that for 75% of successful bids and 40% of unsuccessful bids the agency asked for additional information.

   a. What is the prior probability of the bid being successful (that is, prior to the request for additional information).

   b. What is the conditional probability of a request for additional information given that the bid will be ultimately successful.

   c. Compute the posterior probability that the bid will be successful given a request for additional information.